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ARTICLE

The effects of AI-guided individualized language learning: A meta-analysis

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Abstract

Artificial intelligence (AI) has considerably advanced the methods for individualizing language learning opportunities, such as assessing learning progress and recommending effective individual instruction. In the present study, we conducted a meta-analysis to synthesize recent empirical findings pertaining to the utilization of AI-guided language learning and collected 61 samples (N = 8,282) from 17 research projects (e.g., Assessment to Instruction [A2i], Duolingo, and Project LISTEN). The results of our meta-analysis confirmed that AI-guided individualized language learning was effective for learners' language development (d = 1.18, based on 26 within-group samples, N = 2,262) and had an overall positive treatment effect compared to business-as-usual conditions (d = 0.39, based on 35 between-group samples, N = 6,020). Moreover, the results of our moderator analyses for the treatment effect revealed that AI-guided language learning with machine learning and hybrid systems were more impactful than those with rule-based systems, which may be more helpful (compared to the former) in understanding how predictions are made from a pedagogical perspective. Evidence-based implications are provided based on the results of this meta-analysis.

Keywords: *Artificial Intelligence (AI), Individualized Instruction, Language Learning, Meta-Analysis*

Language(s) Learned in This Study: *English*

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Introduction

Artificial Intelligence (AI) has become common in our daily lives; even smartphones are equipped with AI assistants (e.g., Siri and Google Assistant). In addition, the growth and proliferation of the Chat Generative Pretrained Transformer (ChatGPT) developed by OpenAI (Ghumra, 2022) at an extremely fast pace have accelerated the introduction of AI to the public (Lund & Wang, 2023). Consequently, defining AI is no longer as complex as it was once. Among the various perspectives on how to define AI in the context of education (e.g., Chen et al., 2020; Pokrivcakova, 2019; Sharma et al., 2019), in the present study, we defined AI as "intelligence or machine ability to demonstrate some level of intelligence and perform a wide range of functions and capabilities that require human-like abilities" (p. 75267), borrowing Chen et al.'s (2020) definition in the education field.

AI has been integrated into myriad domains (Xu et al., 2021), with the education field not being an exception (e.g., Du Boulay, 2016; Florea & Radu, 2019; Kandlhofer et al., 2016). The education community is now even raising concerns about AI's potential to replace human teachers, although extant studies have revealed some skepticism regarding this possibility among in-service teachers (e.g., Chan & Tsi, 2023; Tao et al., 2019). Despite this skepticism, statistics suggest that AI will play a larger role in education in the future (GMI, 2022). In the field of language teaching and learning, the potential of AI has been extensively explored in the area of Computer Assisted Language Learning (CALL) (Kessler, 2023).

In the present study, we synthesized AI's pedagogical contributions to supporting individualized language learning. The idea of providing individualized learning to different groups of students in the same classroom

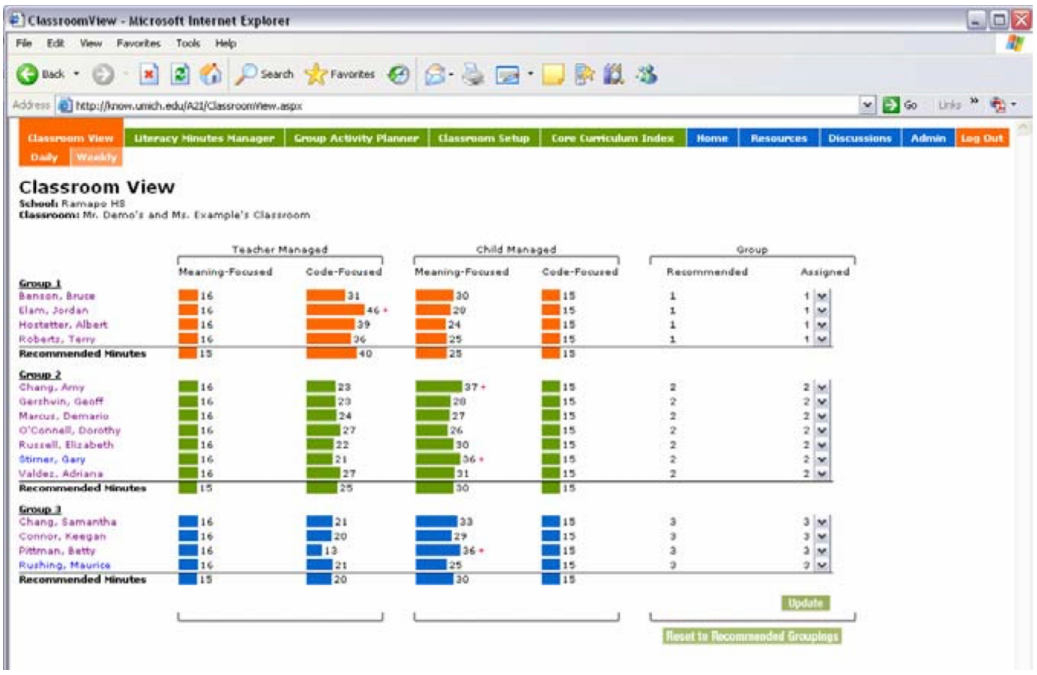
(i.e., differentiated instruction) dates back to the early 20th century (Washburne, 1925). Most teachers would agree that identifying their learners' differences is an important issue in language learning and that being able to respond to such diversity is an integral aspect of high-quality language learning (Puzio et al., 2020). However, this is not an easy task. For example, according to a national survey in the US, 84% of teachers found it difficult to implement such teaching methods (Loveless et al., 2008). Puzio et al.'s (2020) meta-analysis revealed that the overall effect of differentiated literacy instruction was marginal ($d = 0.13$), leaving the potential for such instruction questionable.

The recent growth of AI has made this pedagogical approach much more effective and efficient (Chen et al., 2020), and this contribution has been proposed as "the greatest opportunity for AI in education" (Lee & Qiufan, 2021, p. 118). Thus, we strongly believe that AI-guided individualized language learning has a more promising pedagogical effect on improving individual learners' achievement than Puzio et al.'s (2020) results. Hence, the purpose of the present study was to synthesize the empirical findings on AI-guided individualized language learning opportunities through a meta-analytic lens.

It is critical to point out that "most AIEd [AI in Education] tools have a tendency to drive the homogenization of students" (Holmes & Tuomi, 2022, p. 559), even though many AI-guided applications claim to offer individualized language learning opportunities. Therefore, in this study, we sought to narrow our interpretation of individualized language learning by considering the following two distinct branches of AI technology in the language learning domain. First, one branch developed an AI-based learning program providing learners with individualized pathways by adjusting target instructional approaches (i.e., individualized instruction). Connor et al.'s A2i framework (e.g., 2007, 2009, 2011a, 2011b, 2013) is a relevant example in which the results of individual students' performance through formative assessments are taken into account in the AI algorithm, which in turn recommends different types of instruction combinations for individual students (see Figure 1, where the A2i software suggests the amount of instruction recommended for individual students based on their vocabulary and word reading scores).

Figure 1

A2i Software's Classroom View (Adapted from Connor et al., 2007)



The second branch offers instructional platforms where learners are given individualized paths through different target language items based on their progress (i.e., responses to previous items). A well-known

example in this category is Mostow et al.s' (2003a, 2003b, 2013) Project LISTEN that provides reading tutors, an Intelligent Tutoring System (ITS), with assessment and planning tools as well as performance support, such that learners can improve their oral reading taught in their reading program. Furthermore, Duolingo (e.g., Al Otaiba et al., 2011; Purwanto et al., 2022), a language learning platform founded in 2011, is another relevant example.

It is worth noting that we did not consider AI-based technologies such as chatbots, social robots, and automatic writing experience (AWE), which have already been extensively studied, as evidenced by existing reviews (e.g., Huang et al., 2022; H. Lee & J. H. Lee, 2022a; Nunes et al., 2022).

Background

Individualization for Language Learning

The notion of offering individualized learning opportunities has spawned diverse types of programs in academic contexts (Wang & Walberg, 1985). The field of language learning has also valued the importance of differentiated instruction (International Reading Association, 2000), with diverse programs developed to this end (e.g., School Wide Enrichment Model in Reading, Reis et al., 2011; Individualized Student Instruction, Connor et al., 2007; Integrated Curriculum Model, VanTassel-Baska & Wood, 2010). Theoretically underpinned in Vygotsky's zone of proximal development (1978), individualized instruction is intended to accurately identify a particular student's current stage and then provide a more finely tuned learning opportunity than its one-size-fits-all counterpart. In this way, learning can be "personalized, matched, or adapted to the experiences, aptitudes, and interests of each individual student" (Waxman et al., 2013, p. 405).

It has been proposed that individualized learning will be greatly enhanced by the integration of AI-guided technologies (Lee & Qiufan, 2021), which provide a personalized learning path for each student by analyzing data related to the individual's learning behavior, performance, and patterns. Under this line of thinking, according to Chassignol et al.'s (2018) narrative overview of AIEd, AI has been incorporated into three areas, namely (a) administration, (b) instruction, and (c) learning. Language teaching and learning communities have also recently adopted AI in their respective domains. For example, when borrowing Chassignol et al.'s scope, *Alphie's Alley* (e.g., Chambers et al., 2008a, 2008b), an ITS system for language learning, covers three areas: (a) assessing and providing feedback (administration), (b) determining individualized instructional plans (instruction), and (c) detecting students' learning status and addressing their deficiencies (learning).

Despite such hype for AI-guided technologies for individualized instruction, Holmes and Tuomi (2022) raised some concerns, two of which are discussed below. First, unlike what avid supporters of AI-guided technologies in the education domain claim, individualized instruction can be, and has been given by, human teachers without the aid of such technologies, not to mention no strong evidence of whether the former is more effective than the latter. Second, the individualized learning pathways suggested by AI-guided technologies may not be useful for current students, because such pathways are developed based on prior learners. Given these issues, it is important to examine the extent to which AI-guided individualized learning can benefit language learners' linguistic development.

Only a handful of meta-analyses have synthesized empirical findings related to AI-guided learning (e.g., Lin et al., 2022; Martínez et al., 2023; Zheng et al., 2023), among these it is worth reviewing Lin et al.'s (2022) meta-analysis, which provided a glimpse of the effect of AI on language learning. Although they did not explore the concept of individualization, nine of the 21 empirical studies focused on language learning. Their results revealed that the application of AI in language learning had a small effect size (d = 0.26). As the main objective of this meta-analysis was to provide comprehensive perspectives towards AIEd, it has a limitation that we aimed to address in the present study. More specifically, no further and deeper findings of our interest, such as any moderation effects (i.e., how different features of studies or learners influence the effectiveness of AI in language learning), could be found in any of the previous meta-analysis

studies, including Lin et al. (2022). We aimed to investigate the roles of the following moderator variables particularly significant in employing AI technologies in language learning contexts.

Potential Moderators

This section introduces the five identified moderators hypothesized to influence the effectiveness of AI-guided individualized language learning.

Age

Language learning research has proposed that different age groups harness different learning mechanisms (Bley-Vroman, 1990, 2009) and differ in terms of language learning experience or expertise (Johnson, 2005). Such differences between different age groups could moderate the effects of a particular type of instruction or learning program (e.g., Johnson & Swain, 1994; Lee & Macaro, 2013). Likewise, it can be expected that the effects of AI-guided individualized language learning may vary across different age groups. We hypothesized that younger age groups with the limited learning experience and less developed cognitive maturity would benefit more from the personalized learning paths offered by AI tools because of their lower levels of language learning experience and cognitive maturity than their older counterparts.

Learning Context

AI-guided individualized language learning has been examined in different learning contexts, including first, second, and foreign language ones. In the case of the first language learning, in which differentiated instruction is largely associated with literacy development (International Reading Association, 2000), AI-guided individualized language learning has also been oriented toward enhancing literacy (i.e., reading skills) (e.g., Chambers et al., 2008a, 2008b, 2011; Connor et al., 2007, 2009, 2011a, 2011b; Mills-Tettey et al., 2009; Mostow et al., 2013). Another group of studies (e.g., Ali, 2021; Aulia et al., 2020; Kusumadewi & Widyastuti, 2018; Nagata, 1993; Purwanto et al., 2022) has been conducted in foreign language contexts, in which the target language is not spoken outside the classroom, and motivation to learn the target language is often career-oriented (Dewaele, 2009). Finally, some studies (e.g., Heilman et al., 2010; Loewen et al., 2020; Poulsen, 2004; Vesselinov & Grego, 2012) have targeted second language learners, for whom the target language is the language spoken in the community of their current residence. Given the range of differences between contexts (e.g., the motivation to learn the target language, amount of available input outside the classroom, and status of the target language), we hypothesized that context may be an important moderator that could influence the effects of AI-guided individualized language learning.

Target Language Domain

As mentioned previously, individualized instruction has been largely associated with literacy development. Thus, it is expected that a large proportion of research on AI-guided individualized instruction has set reading (i.e., literacy) as the target language domain and has examined the effects of the researchers' programs on literacy development. These programs include Chambers et al.'s (2008a, 2008b, 2011) Alphie's Alley, Connor et al.'s (2007, 2009, 2011a, 2011b, 2013) ISI with A2i software, and Mostow et al.'s (2003a, 2003b, 2013) Project LISTEN Reading Tutor. Apart from reading, vocabulary has also been targeted extensively in research on AI-guided individualized instruction, presumably because of "the characteristics of vocabulary lessons, in which the target (lexical) items could be presented individually and independently" (H. Lee & J. H. Lee, 2022a, p. 121). Examples include Duolingo (e.g., Ali, 2021; Aulia et al., 2020; Kusumadewi & Widyastuti, 2018; Zamzami, 2019) and Memerise (e.g., Esmaeili & Shahrokhi, 2020; Nuralisah & Kareviati, 2020; Rohim et al., 2022). Finally, other skills such as listening, speaking, and grammar have been targeted in a limited number of studies (e.g., Hassani et al., 2016; Loewen et al., 2020; Maesaroh, 2021; Purwanto et al., 2022; Tsiriga & Virvou, 2004).

AI Assistance Role

When AI technologies are adopted in empirical studies, their roles vary depending on the objectives of the study. According to Lin et al. (2022), for example, approximately 78% of the studies collected for their

meta-analysis employed ITS to provide their students with virtual tutors for individualized language learning (c.f., the remaining 22% were not relevant to our definition of individualized language learning). In the case of Zheng et al. (2023), where a total of seven different roles were identified for their moderator analyses, (a) ITS, (b) testing and diagnostic systems/assessment or monitoring systems, and (c) adaptive learning systems (ALS) / learning support tools were relevant to AI-guided individualized language learning. Similarly, although Hwang's (2022) meta-analysis focused on mathematics, this study found that the majority (approximately 63%) of previous AI-guided instructions employed either (a) ITS or (b) ALS. Others have employed language learning tools to provide instructions to individual learners according to their learning levels and pace using AI systems (see Godwin-Jones, 2021 for a recent trend of using big data in language learning).

AI System Type

In general, AI accepts input data (e.g., text, speech, images), processes the data according to various methods (e.g., rule-based algorithm systems that apply manually specified rules to solve problems, machine learning algorithm systems that make predictions based on automatically learned rules), and provides outcomes such as predictions or classifications (see Lee et al., 2019a for an exemplary language learning study). Depending on how the AI is designed, particularly the types of algorithms (i.e., rule-based algorithm systems, machine learning algorithm systems, or both), the results can be analyzed and used to adjust the system and then repeat the entire procedure (i.e., loop) until the desired end state is achieved (Witten et al., 2016). One example of a rule-based AI system is Connor et al.'s A2i framework (e.g., 2007, 2009, 2011a, 2011b), whose developed AI algorithm was modeled after instructors' classroom management, and the AI system analyzed assessment data to provide ideal recommendations based on pre-populated rules. Considering machine learning AI systems, such as Duolingo (e.g., Ali, 2021; Aulia et al., 2020; Esmaeili & Shahrokhi, 2020), the AI algorithm determines the most appropriate lessons based on students' specific needs. The algorithm then learns to improve its accuracy in response to student interactions. Given that the number of users in the third quarter of 2022 was approximately 56.5 million (company data), improving the accuracy of Duolingo's AI algorithm would have been unrealistic. Therefore, different types of AI systems may influence the overall effectiveness of their employment for individual students. Although previous meta-analyses (e.g., Hwang, 2022; Lin et al., 2022; Zheng et al. 2023) have identified the different roles of AI employed by previous empirical studies, none of them dealt with the types of AI systems or algorithms.

The Present Study

The goal of the present study was to synthesize the empirical findings of AI-guided individualized instructions for language learning, through a meta-analytic lens. To obtain a more balanced view of the effects of AI-guided individualized instructions, we identified both the within-group (i.e., pre-post design) and between-group studies (i.e., those that compare the experimental and business-as-usual groups). The following three research questions guided the current meta-analysis to achieve this research goal.

1. To what extent is AI-guided individualized learning effective for language learning?

2. To what extent does AI-guided individualized language learning have a significant treatment effect compared to the business-as-usual condition?

3. What are the important learner and contextual factors influencing the treatment effects of AI-guided individualized language learning? To what extent do AI's various roles and algorithms relate to its treatment effects?

Methods

Literature Search

To initiate the meta-analysis, we conducted a literature search after defining the following four inclusion

criteria. To be included in the current meta-analysis, a study must (a) be designed to improve learners' language learning achievements, (b) adopt AI to support individualized language learning, (c) report sufficient descriptive statistics to calculate Cohen's *d* for effect sizes (e.g., means and standard deviations for test results and sample sizes), and (d) be written in English. To identify as many studies as possible, we did not restrict the publication years or article types. The authors of the present study conducted a literature search, which consisted of a database, journal, and backward/forward reference searches.

First, we conducted a database search. For this, we visited Web of Science (WoS), ProQuest, and SCOPUS and used the following keyword combination in the title and abstract fields in accordance with four key parameters: (a) instructional types: intelligent tutoring system ("intellige* tutor*" OR "intellig* agent" OR "cognit* tutor*" OR "adapt* tutor*" OR "cognit* virtual companion*" OR "intellige* coaching system*" OR "adapt* instruct*" OR "adapt* teach*"), differentiated instructions ("differentiat* instruct*" OR "differentiat* teach*"), or individualized instructions ("individuali* instruct*" OR "individuali* teach*"); (b) language learning topics: first language learning ("language learning" OR "language teaching" OR "language acquisition"), second/foreign language learning ("second language" OR "foreign language" OR L2), or English learning ("English learning" OR "English teaching" OR "English acquisition" OR EFL OR ESL); (c) adoption of AI systems (CALL OR agent OR interface OR "artificial intelligence" OR AI OR "computer-assisted" OR "computer*"); and (d) quantitativeness ("impact*" OR "effect*"). Thus, we identified relevant primary studies in the current meta-analysis.

For journal searches, we visited relevant academic journals, including *Annual Review of Applied Linguistics, CALICO Journal, Computer-Assisted Language Learning, Educational Research Review, Language Learning, Language Learning & Technology, Language Teaching Research, ReCALL, Review of Educational Research, Studies in Second Language Acquisition,* and *TESOL Quarterly*. The aforementioned keywords were used again for the journal searches.

After identifying relevant literature on the topic, we conducted backward and forward reference searches. For the backward reference search, we checked the reference lists of the identified studies; for the forward, we used Google Scholar's *Cited By* function to locate follow-up studies that cited the identified studies. A total of 186 cases were identified after removing duplicates.

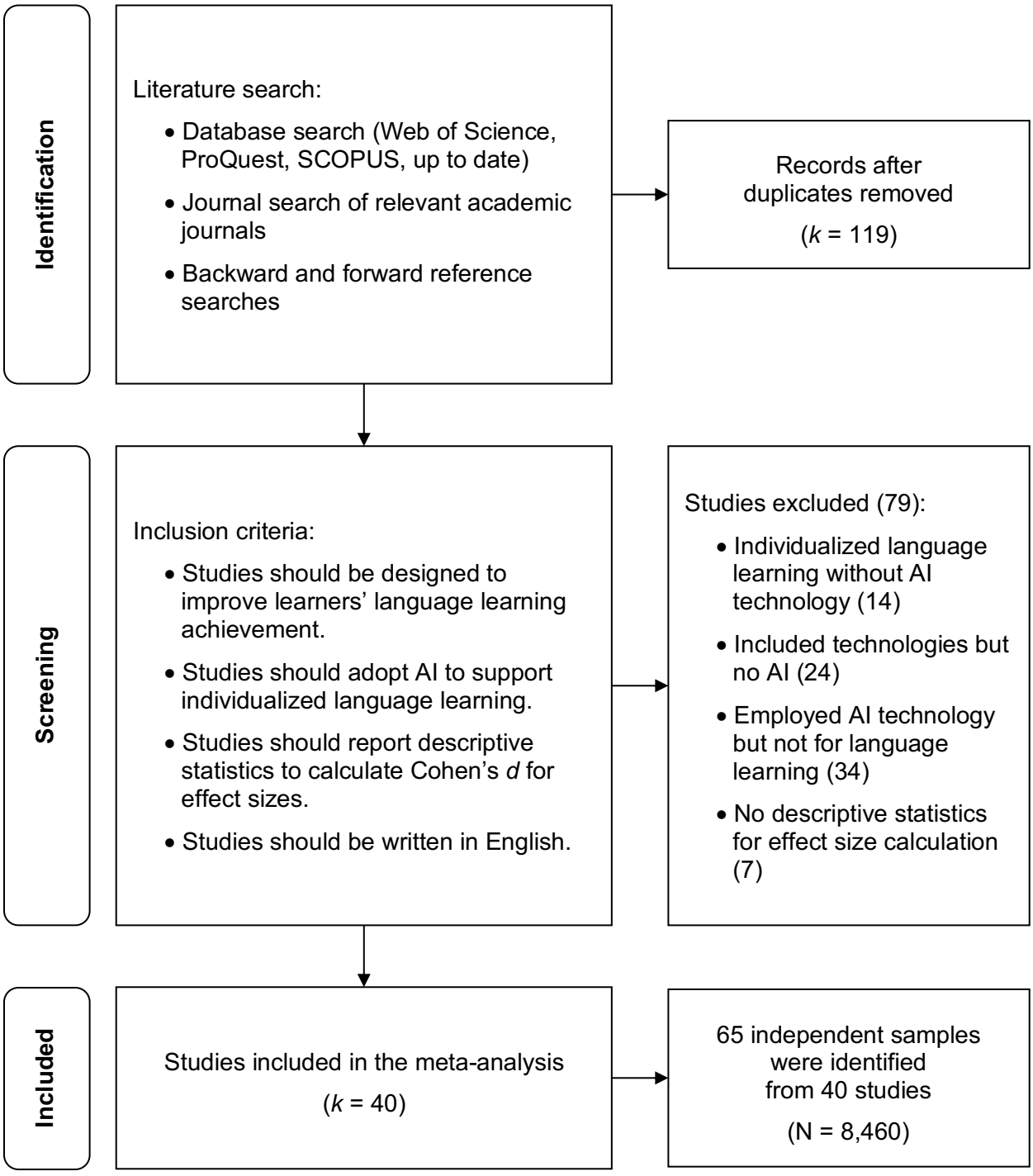
The titles and abstracts of these 186 cases were then reviewed for their relevance to the topic, and 119 cases remained eligible for a full review to verify whether they successfully met the four inclusion criteria. As shown in Figure 2, a total of 79 studies were excluded for the following reasons: (a) studies included individualized instructions (e.g., differentiated instructions, one-on-one tutoring) but AI technologies were not employed (k = 14); (b) studies used diverse computer technologies (e.g., web-conferencing system, online coaching, computerized formative assessment), but AI technologies were not included (k = 24); (c) studies leveraged AI technologies but did not measure language learning outcomes (k = 34); and (d) studies did not report descriptive data for effect size calculations (k = 7). As a result, 40 studies were selected for the current meta-analysis (see Supplementary Material for the table outlining each study's information).

Dataset Construction

This section delineates the entire procedure for constructing datasets: (a) identifying independent samples and their multilevel data structures, (b) estimating effect sizes, (c) detecting outliers and influential cases and checking any sign of small-study effects, and (d) coding moderators. As we included both one-group (i.e., pre-post design) and two-group experimental studies (i.e., experimental group vs. business-as-usual group), there were two types of datasets: one for within-group effect sizes and the other for between-group effect sizes. All the procedures were performed by the authors of the current study. Any disagreements were resolved through productive discussions.

Figure 2

Search and Exclusion Process (PRISMA flow diagram)



Understanding Data Structure

The identified 40 studies were then reviewed to understand the structure of the dataset. First, some of the studies used multiple independent samples. For example, Chambers et al. (2011) included both first- and second-graders and reported results separately for each population, thus representing two independent

samples (as if there were two distinguishable studies; thus, we refer to them as Chambers et al., 2011[1] and Chambers et al., 2011[2]). Similarly, Mills-Tettey et al. (2009) recruited three different populations from (a) a private school, (b) a public school, and (c) an informal educational program and reported separate results for each population. Therefore, we identified three samples from this study, referred to as Mills-Tettey et al. (2009) [1], Mills-Tettey et al. (2009) [2], and Mills-Tettey et al. (2009) [3]. As a result, we identified a total of 65 samples (N = 8,460), consisting of 26 pre-post (i.e., within-group; differences within treatment groups [between pre-test and post-test] with no business-as-usual groups) samples (N = 2,262) and 39 between-group (i.e., differences between treatment and business-as-usual groups [between post-test results]) samples (N = 6,198).

Second, some studies were conducted by the same or nearly the same researchers as part of a larger research project. Chambers et al. (2008a), Chambers et al. (2008b), Chambers et al. (2011) [1], and Chambers et al. (2011) [2] all used the same intelligent tutoring system (ITS), called Alphie's Alley. Similarly, Ali (2021), Aulia et al. (2020), Kusumadewi and Widyastuti (2018), Purwanto et al. (2022) [1], Purwanto et al. (2022) [2], and Zamzami (2019) employed Duolingo. If the same number of participants were measured in multiple studies, the most recent studies were included in the meta-analysis. However, none measured the same participants multiple times (i.e., each study had independent samples). Thus, we conclude that our 65 samples were nested in 17 research projects.

Estimating Effect Sizes

Effect size estimation is key for meta-analyses, as it allows to mathematically compare empirical findings from different studies with different outcome measures. One of the most common approaches is to compute the standardized mean difference (SMD), also known as Cohen's d, by dividing the mean difference for each outcome between two groups by the pooled standard deviation between the two groups. More precisely, pre-test and post-test results are considered for one-group design studies, and treatment and business-as-usual groups' post-test results in the case of two-group experimental design studies. Thus, standardized effect sizes have the advantage of being numeric scales for measuring the strength of the treatment effects. Therefore, we can objectively synthesize the magnitudes of the effects of AI-guided individualized language learning among different empirical studies. During the estimation, given that there were some studies with small sample sizes (<50), we added *J correction factor* to the effect size calculation to avoid possible bias (Hedges, 1981); the unit of our effect sizes was *unbiased* Cohen's d (also known as Hedges' g). After estimating the effect sizes for each outcome, we computed 119 effect sizes.

Detecting Outliers/Influential Cases and Publication Bias

After the effect size calculation, we conducted outlier/influential case diagnostics using the *metafor* package (Viechtbauer, 2010) in the R software. This approach quantifies how certain cases are considered outliers and influential cases based on several mathematical indices, such as studentized deleted residuals, DFFITS, Cook's distance, and COVRATIO (see Viechtbauer & Cheung, 2010 for their definitions). Based on systematic diagnostics quantified systematically, this approach mathematically detects outliers and influential cases whose exclusion may considerably impact the overall meta-analysis. Thus, outliers should be excluded, whereas a follow-up sensitivity analysis is needed for influential cases to check whether the inclusion or exclusion of influential cases may have affected the results of the meta-analysis (see Viechtbauer & Cheung, 2010 for the rationale and J. H. Lee & H. Lee, 2022b for a practical guideline). The results indicated that none of the within-group samples were detected as outliers; however, four effect sizes were identified in the pool of between-group samples (i.e., no influential case was detected for both datasets). We, therefore, excluded the four potential outliers in the related pool. As a result, 115 effect sizes from 61 samples (N = 8,282) based on 17 projects remained, that is, 39 effect sizes from 26 within-group samples (N = 2,262) based on 14 projects and 76 effect sizes from 35 between-group samples (N = 6,020) based on 13 projects. Tables 1 and 2 present the structures of the datasets.

After detecting the outliers, we checked for signs of publication bias. Considering that studies with statistical significance tend to be published more frequently, synthesizing published primary studies for

meta-analysis may neglect unpublished studies with null findings or negative results (Song et al., 2010), which are likely underrepresented in the literature. More specifically, if the calculated effect sizes from the collected studies are predominately in one direction, this may indicate publication bias, suggesting that the dataset may canonize false facts. A funnel plot is a widely used approach for visualizing the symmetrical distribution around effect sizes and standard errors (Debray et al., 2018). Egger's regression test (Egger et al., 1997) is another common method for statistically checking publication bias.

Table 1

Entire Structure of the Within-Group Sample Dataset

Level 3: Project (14)	Level 2: Sample (26)	Level 1: Outcome (39)
Alphie's Alley	Chambers et al. (2008a)	1
Autotutor Software	Mohammadzadeh & Sarkhosh (2018) [1]	1
	Mohammadzadeh & Sarkhosh (2018) [2]	1
Babbel	Loewen et al. (2020)	3
Bussu	Maesaroh (2021)	1
Duolingo	Ali (2021)	1
	Vesselinov & Grego (2012)	1
	Purwanto et al. (2022) [1]	1
	Purwanto et al. (2022) [2]	1
ISI with A2i	Al Otaiba et al. (2011)	3
	Connor et al. (2007)	3
	Connor et al. (2009)	3
	Connor et al. (2011a)	2
	Connor et al. (2011b)	2
iSTART-E	McCarthy et al. (2020)	1
ITSS	Wijekumar et al. (2012)	1
IVELL	Hassani et al. (2016)	4
Memrise	Esmaeili & Shahrokhi (2020)	1
	Nuralisah & Kareviati (2020)	1
	Rohim et al. (2022)	1
	Taebenu & Katemba (2021) [1]	1
	Taebenu & Katemba (2021) [2]	1
Moodle	Jia et al. (2012)	1
Negative Knowledge	Demir (2020)	1
Nihongo-CALI	Nagata (1995)	1
Web-based ICALL	Tsiriga & Virvou (2004)	1

Table 2

Entire Structure of the Between-Group Sample Dataset

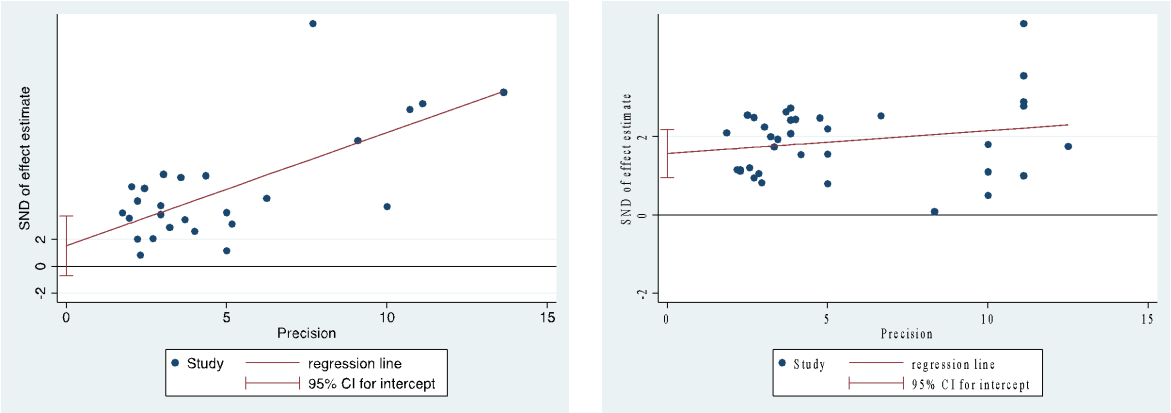
Level 3: Project (13)	Level 2: Sample (35)	Level 1: Outcome (76)
Alphie's Alley	Chambers et al. (2008a)	4
	Chambers et al. (2008b)	3
	Chambers et al. (2011) [1]	3
	Chambers et al. (2011) [2]	2
Duilingo	Ali (2021)	1
	Aulia et al. (2020)	1
	Purwanto et al. (2022) [1]	1
	Purwanto et al. (2022) [2]	1
ISI with A2i software	Al Otaiba et al. (2011)	5
	Connor et al. (2007)	3
	Connor et al. (2011a)	2
	Connor et al. (2011b)	2
	Connor et al. (2013) [1]	1
	Connor et al. (2013) [2]	1
	Connor et al. (2013) [3]	1
iSTART	McNamara et al. (2006)	2
ITSS	Wijekumar et al. (2012)	1
Memrise	Esmaeili & Shahrokhi (2020)	1
	Rohim et al. (2022)	1
Moodle	Jia et al. (2012)	1
Negative Knowledge	Demir (2020)	1
Nihongo-CALI	Nagata (1993)	1
	Nagata (1995)	1
	Nagata (1996)	1
Project LISTEN	Mostow et al. (2003a)	2
	Mostow et al. (2003b)	1
	Mostow et al. (2013)	12
	Poulsen (2004)	5
	Aist et al. (2001) [1]	1
	Mills-Tettey et al. (2009) [1]	2
	Mills-Tettey et al. (2009) [2]	2
	Mills-Tettey et al. (2009) [3]	2
REAP	Heilman et al. (2010)	3
The First 400 Words	Fehr et al. (2012)	3
Web-based ICALL	Tsiriga & Virvou (2004)	1

Egger's test results revealed that the calculated effect sizes for the within-group samples did not show signs of publication bias ($\beta = 1.53$, $SE = 1.08$, $p = .17$), whereas those for the between-group samples showed a statistically significant sign of bias ($\beta = 1.56$, $SE = .30$, $p < .001$) (see Figure 3). During the literature search phase, we located grey literature (e.g., conference papers and research reports) as thoroughly as possible, but our efforts did not remedy this issue. Nevertheless, we identified four control variables: (a) duration of the instruction (in weeks), (b) publication year (mean centered), (c) study design (i.e., randomized controlled trial vs. quasi-experimental design study), and (d) test reliability (i.e., standardized test with reported reliability vs. researcher developed test without reliability reported), which concern publication types of the primary studies. We included them in our meta-regression models to check for publication bias effects (H. Lee et al., 2022; Peng et al., 2021). The results showed no statistical significance, indicating that publication types in the dataset would not influence the overall results. However, given that one of the two

datasets (i.e., the between-group samples) possibly had a publication bias, we refrained from making a definitive argument when interpreting the related results.

Figure 3

Results of Egger's Test for Within-Group (left) and Between-Group (right) Samples



Coding Moderator Variables

Having checked for publication bias, we then identified quantifiable features that can categorize how the collected primary studies differed from each other. Statistically controlling for these features (i.e., moderator variables) would provide a more reliable perspective toward synthesizing the computed effect sizes from diverse settings (Woltman et al., 2012). As discussed above, we identified five moderators: (a) students' age, (b) language learning contexts, (c) target language domains, (d) AI assistance roles, and (e) AI system types, all of which may influence the impact of AI-guided individualized language learning. Table 3 presents the coded moderator variables.

First, the studies included participants of various ages. Considering the American Medical Association's age designations (NIH, 2022), we categorized the participants into three groups: children (one year through 12 years), adolescents (13 years through 17 years), and adults (18 years or older).

Second, there were three types of language learning contexts: first-language learning (e.g., learning English in the USA), second-language learning (e.g., learning Spanish in the USA), and foreign-language learning (e.g., learning English in Indonesia).

Third, the foci of the target language domain were diverse, although a large proportion of the effect sizes were associated with reading skills and vocabulary knowledge, and a smaller number of effect sizes were associated with speaking skills, listening skills, grammatical knowledge, and overall language skills. During the meta-regression analysis phase, we found that we could not keep all the skills individually for this moderator, as it made regression models inestimable due to a lack of statistical power (i.e., too many categories but too small samples). Thus, we had to lump some of the values in accordance with their sample sizes. Moreover, we found that the pool of between-group effect sizes did not include relevant samples for "speaking skills" and "overall language skills." Therefore, we designated three categorical values: (a) reading skills, (b) vocabulary knowledge, and (c) others. The last categorical value included the cases of "grammar knowledge," "speaking skills," "listening skills," and "overall language skills."

Fourth, the role of AI systems was threefold, such that some studies employed ITS based on AI systems to provide individualized instructions and feedback (see Ma et al., 2014 for a review). Another set of studies used AI systems to help teachers analyze their students' assessment data and recommend a suitable amount and type of literacy instruction for each student (i.e., learning management; see Deunk et al., 2018 for a general view of the individualized learning management approach). The last type included individual

language learning tools equipped with adaptive learning systems, such as Duolingo.

Finally, the types of AI systems used in each primary study were diverse. After reviewing the primary studies collected for the meta-analysis, we found that whether the AI system was designed to adjust its rules and algorithms manually or automatically constituted one of the most distinguishable features among the AI systems adopted by the primary studies. Regarding other AI system-related features, such as their input data formats, data processing methods, or outcome types, the scope of variations could not be categorized to objectively compare them from both theoretical and practical perspectives (Witten et al., 2016). Consequently, we coded the AI system types into three values: (a) rule-based systems (i.e., no adjustment made after each loop), (b) machine learning systems (i.e., the intelligence of a machine enables a system adjustment to be made after each loop until the desired goals are met), and (c) hybrid systems (i.e., human programmers manually adjust the system based on the results).

Table 3

Description of the Coded Moderator Variables

Moderator variable	# of values in between-group samples (k = 35)
Age	
Children	22
Adolescents	4
Adults	9
Learning context	
First language learning	21
Second language learning	2
Foreign language learning	12
Target language domain	
Reading	20
Vocabulary	8
Others (Grammar)	5
(Speaking)	0
(Listening)	2
(Overall language skills)	0
AI assistance role	
Intelligent tutoring system	20
Learning management	7
Individual learning tool	8
AI system type	
Rule-based system	21
Machine learning system	6
Hybrid system	8

Dataset Analysis

After calculating the effect sizes and finishing the coding procedure, subsequent data analyses consisted of two stages: (a) synthesizing the calculated effect sizes to compute the overall average effect sizes and (b) estimating how the identified moderators influenced the overall average effects. For both phases, we used STATA version 16 (StataCorp, 2019) to employ a multilevel meta-analysis approach using a random-effects model. Given that the datasets had a three-level structure (level 1: outcome; level 2: sample; level 3: project) and that the effect sizes (at level 1) were estimated based on outcomes derived from different populations (i.e., subject to standard errors), estimating a three-level regression with a known variance model was our approach (see Hox et al., 2018 for the theoretical background; see H. Lee et al., 2019b, H. Lee et al., 2020 for practical guidelines and examples). Thus, in the first stage, we estimated a baseline regression model to identify the average effect sizes of the two datasets. Given that the intercept of a

baseline model (i.e., with no independent variables) is a parameter for the entire effect size, the intercept value is the overall average effect size (Raudenbush & Bryk, 2002).

In the second stage, we investigated the moderation effects for the second dataset (i.e., 35 between-group samples). When moderators are included as independent variables in the baseline model, all the model's coefficients are interpreted as differences between the modeled and reference values (Raudenbush & Bryk, 2002; Woltman et al., 2012). Thus, we included each moderator variable in separate regression models (i.e., simple regression models) to compute the average effect sizes for each categorical value of the moderators (i.e., subgroup analysis). Next, every moderator variable was included in the baseline model to investigate the overall moderation effects (i.e., meta-regression; Borenstein et al., 2021) after controlling for each moderator, along with the aforementioned four control variables related to publication types. When we identified any moderator variables with statistical significance at this stage, we conducted another meta-regression analysis using those significant variables to obtain more accurate and precise estimates (Hox et al., 2018). As the results of the meta-regression are generally considered more accurate than those of the subgroup analysis, we included the latter in Appendix owing to space limitations.

Results

Research Question #1. To what extent is AI-guided individualized learning effective for language learning?

The first research question was answered based on the within-group sample dataset consisting of 39 effect sizes from 26 within-group samples (N = 2,262), based on 14 projects. These effect sizes were calculated by dividing the mean differences between the pre- and post-test results by their pooled standard deviations. Furthermore, they represent the degree of development language learners achieve through AI-guided individualized language learning. As detailed in the forest plot (Figure 4), the effect sizes ranged from 0.23 to 2.90 across the 26 within-group samples. Although two had zeros in their confidence intervals, indicating that these studies reported the null effects of AI-guided individualized language learning, most samples showed positive contributions (ds > 0) of AI technology on linguistic development.

As Table 4 demonstrates, after synthesizing the effect sizes from these 26 samples, we found that the overall average effect size was 1.18 (SE = .17, p < .001), which suggests that, on average, AI-guided individualized language learning made statistically significant contributions to linguistic development and that the impact was large (d > .80) according to Cohen's effect size interpretation benchmarks.

Table 4

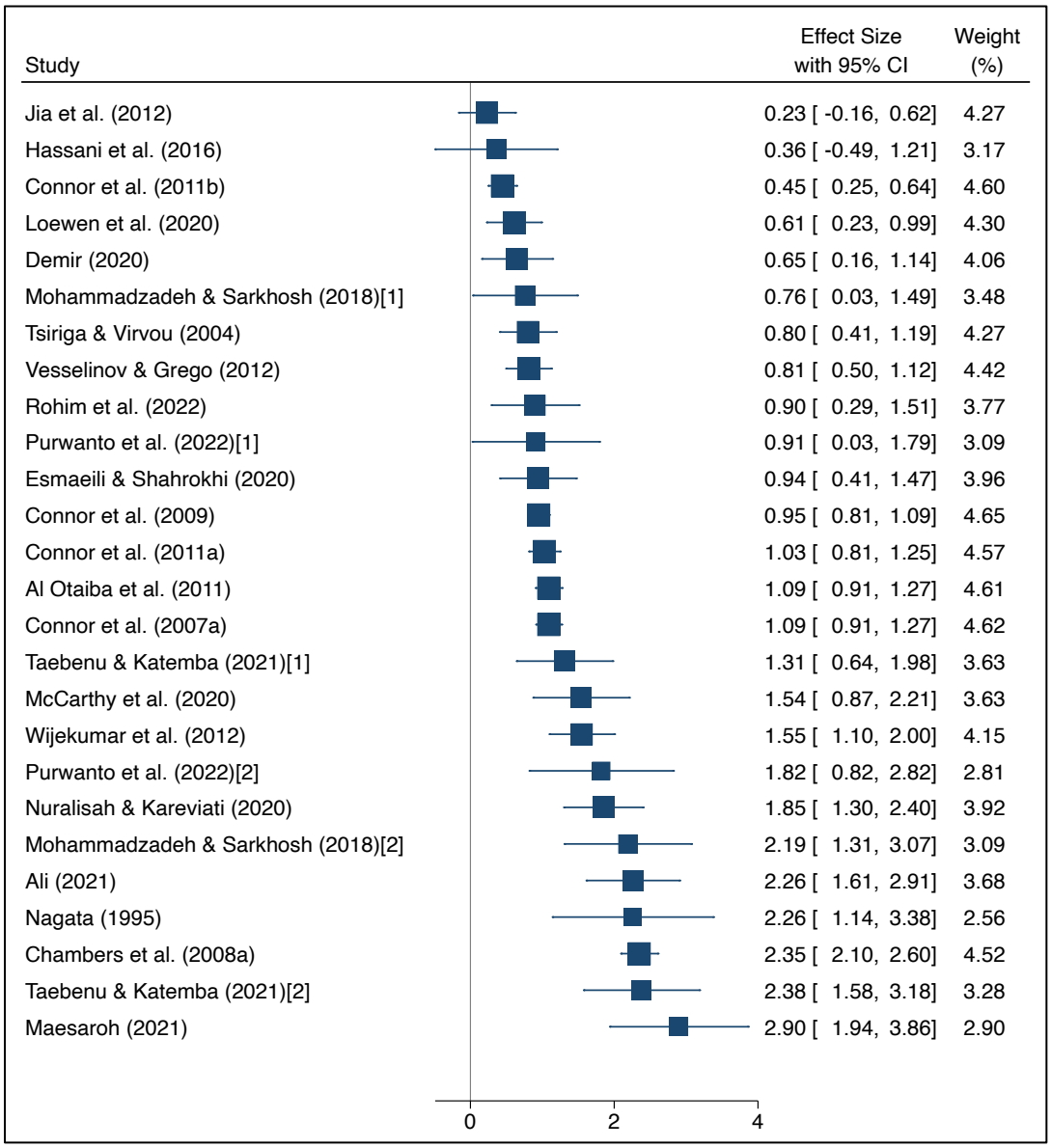
Results of Meta-analysis for Average Within-Group Effect Size (N = 2,262)

# of ES	# of Sample	# of Project	Average ES	SE
39	26	14	1.18***	(0.17)

Note. *** p < .001

Figure 4

Forest Plot for 26 Within-Group Samples



Research Question #2. To what extent does AI-guided individualized language learning have a significant treatment effect compared to the business-as-usual condition?

To answer the second research question, the between-group sample dataset, consisting of 76 effect sizes from 35 samples (N = 6,020) based on 13 projects, was analyzed. The effect sizes calculated from the between-group samples represent the treatment effects of AI-guided individualized language learning compared to business-as-usual conditions, according to the standardized mean differences between the post-test results of the treatment and business-as-usual groups. Therefore, any positive value indicates the superiority of AI-guided language learning over conventional language learning conditions. As detailed in the forest plot (Figure 5), the calculated effect sizes ranged from 0.01 to 1.13, and none of the 35 samples had a statistically significant negative effect size. These outcomes indicate that AI-guided individualized language learning was at least as effective as learning under conventional conditions. Approximately half (17 samples) of the samples had positive effect sizes with statistical significance (ds > 0); this suggests the

possibility that AI-guided individualized language learning may have been more effective in enhancing students' linguistic development than the business-as-usual classroom conditions were.

Figure 5

Forest Plot for 35 Between-Group Samples

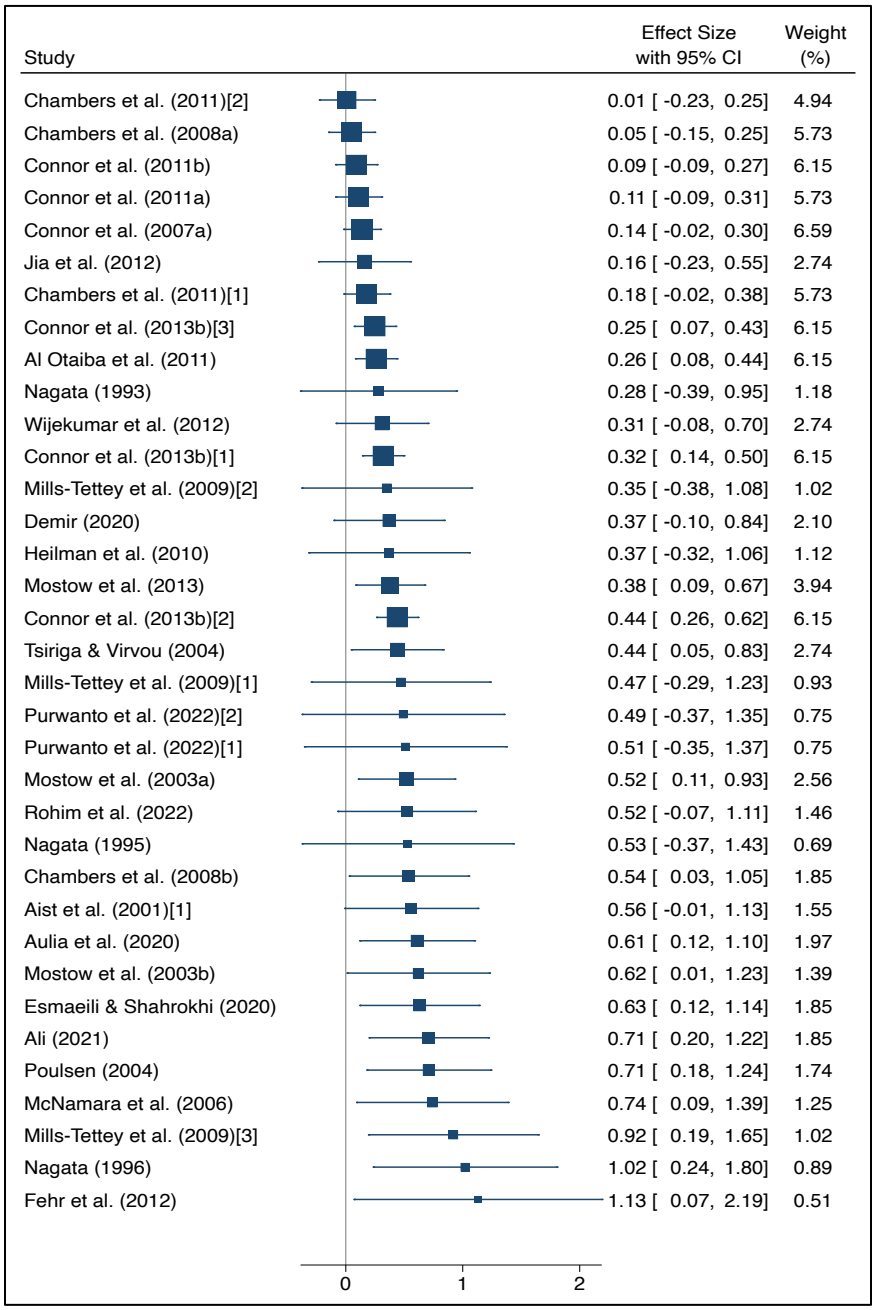


Table 5 presents the overall average of the between-group effect size estimates. The results indicated that the average effect size was 0.39 (SE = .07, p < 001), suggesting that, on average, AI-guided individualized language learning had a significantly higher educational impact than the conventional ones (i.e., a positive treatment effect). Furthermore, the average treatment effect was small (.20 ≤ d. < .50), in accordance with Cohen's d interpretation benchmark.

Table 5

Results of Meta-Analysis for the Average Between-Group Effect Size (N = 6,020)

# of ES	# of Sample	# of Project	Average ES	SE
76	35	13	0.39***	(0.07)

Note. *** $p < .001$

Research Question #3. What are the important learner and contextual factors influencing the treatment effects of AI-guided individualized language learning? To what extent do AI's various roles and algorithms relate to its treatment effects?

To answer the third research question, multiple regression analysis, also known as meta-regression, was conducted by including five moderator variables that aligned with our interests and four control variables related to publication types (see Table 6) to investigate further whether there were any statistically significant differences across categorical values of moderator variables. As mentioned in the Data Analysis section, the results of our subgroup analysis as part of this moderator analysis are included in Appendix.

Table 6

Results of Moderator Analysis (Meta-Regression) for Between-Group Samples (N = 6,020)

Independent variable	Multiple regression analysis (meta-regression)			
	Coefficient	SE	Adj. average ES	SE
Age				
Children	0.00	(.)	0.45***	(0.07)
Adolescents	0.01	(0.26)	0.46*	(0.22)
Adults	-0.40	(0.32)	0.05	(0.27)
Context				
First language learning	0.00	(.)	0.31***	(0.09)
Second language learning	0.01	(0.21)	0.32	(0.19)
Foreign language learning	0.45	(0.40)	0.76*	(0.33)
Target language domain				
Reading	0.00	(.)	0.35***	(0.10)
Vocabulary	0.29	(0.34)	0.63*	(0.26)
Others	0.03	(0.49)	0.37	(0.41)
AI assistance role				
Intelligent tutoring system	0.00	(.)	0.44***	(0.09)
Learning management	0.06	(0.09)	0.50***	(0.11)
Individual learning tool	-0.46	(0.57)	-0.02	(0.50)
AI system type				
Rule-based system	0.00	(.)	0.28***	(0.06)
Machine learning system	0.42	(0.37)	0.70*	(0.34)
Hybrid system	0.23*	(0.10)	0.51***	(0.08)

Note. *** $p < .001$, * $p < .05$

The results indicated that only one moderator variable was statistically significant: the AI system type. Among the three categorical values, AI-guided individualized language learning with hybrid systems had a statistically significant coefficient of 0.23 ($SE = .10$, $p = .017$) when compared with rule-based systems as the reference level.

Table 7

Results of Follow-Up Meta-regression with Significant Moderator for Between-Group Samples (N = 6,020)

Independent variable	Coefficient	SE	Adj. average ES	SE
AI system type				
Rule-based system	0.00	(.)	0.29***	(0.05)
Machine learning system	0.30	(0.24)	0.59**	(0.22)
Hybrid system	0.22**	(0.08)	0.51***	(0.07)

Note. *** $p < .001$, ** $p < .01$

To elicit more precise estimates, we conducted a follow-up meta-regression analysis. The results are shown in Table 7. Similar to the full meta-regression model, AI-guided individualized language learning using hybrid systems had a statistically significant coefficient of 0.22 (*SE* = .08, *p* = .007). Additionally, the machine learning system type had a larger coefficient (β = .30) without statistical significance; however, it did not reach statistical significance because of its large standard error (*SE* = .24), which was presumably due to its relatively small sample size and complex data structure (i.e., six effect sizes nested in six samples from two projects). According to the adjusted average effect size estimates, we identified that both hybrid systems (*d* = 0.51, *SE* = .07) and machine learning systems (*d* = 0.59, *SE* = .22) had medium average effect sizes, whereas rule-based systems had small average effect size (*d* = 0.29, *SE* = .05).

Discussion

This meta-analysis aimed to synthesize the quantitative findings of primary empirical studies that adopted AI-guided individualized language learning to facilitate students' language learning. We contributed to this branch of research in the following two aspects: (a) the overall effectiveness and positive treatment effects of using AI-guided individualized language learning and (b) the directions of how AI-guided language learning systems can be used more effectively. Taken together, AI-guided individualized language instruction, if equipped with well-developed learning curricula and materials (developed by human experts) and highly accurate predictive models (either by pre-populated rules, big data collected by the machine itself, or both), can be utilized more effectively for certain types of learners in specific learning environments. For example, considering learners who have difficulty accessing an appropriate educational environment, such as when schools are so remote that there are few teachers of the target language, this method would be the most effective. The following sections delineate the two primary findings of this study.

First, this meta-analytic study provides robust and summative evidence of the overall effectiveness of AI-guided individualization in the language learning domain. Our results regarding within-group samples revealed that AI-guided individualized language learning could be considered to have a large (i.e., $d \geq .80$) average effect size ($d = 1.18$). This finding is in line with the general premise of sociocultural theory (Bruner, 1983; Vygotsky, 1978), which underlines the importance of gauging each learner's current stage and offering individualized instruction and supports general positivism for integrating AI technologies into education (Lee & Qiufan, 2021). When compared with conventional language learning environments, a small (i.e., $.20 \leq d < .50$) average effect size ($d = 0.39$) was observed for the between-group samples, confirming that AI-guided individualized language learning was at least as effective as learning in conventional conditions.

Thus, the findings of the current study contribute to dispelling existing skepticism about individualization through AI technologies in education (Holmes & Tuomi, 2022). Therefore, policymakers and material developers in the language-learning domain should consider developing and adopting AI-guided individualized learning in their contexts. For example, when designing a learner's reading development curriculum, material developers can refer to existing AI-guided individualized programs (e.g., Connor et

al.'s A2i framework, 2007; 2009; 2011a, 2011b) and incorporate the relevant components into their efforts. In addition, they should constantly explore how AI-guided individualized language learning systems and applications can supplement students' language learning when teachers and classroom resources are scarce.

Second, another contribution of this study comes from the outcomes of the moderator analyses, which allowed us to obtain a deeper understanding of the types of AI technology that could aid individualized language learning the most. Our meta-analysis demonstrated that across the different system types (i.e., rule-based systems, machine learning systems, or hybrids), AI-guided individualized language learning could bring about the largest treatment effect with a hybrid algorithm system that incorporates human involvement in machine learning algorithm systems. Project LISTEN (e.g., Beck et al., 2003; Mostow, 2016; Mostow et al., 2013) is an exemplary AI system that combines rule-based and machine learning systems. Their collected data, including a myriad of words heard by the speech recognizer and students' clicking behaviors, were used to acquire predictive models as part of a machine-learning approach (e.g., Beck & Woolf, 2000). Meanwhile, AI systems was updated by human researchers based on field-testing results (e.g., González-Brenes & Mostow, 2011). Based on this finding, we propose that teachers' expertise and theoretical backgrounds can be used as guiding logic to make decisions based on inferences made by machine learning algorithms from large amounts of learner data because the human component plays a crucial role in the hybrid system, which takes advantage of both systems (Lessware, 2021). In this way, when individualized recommendations for unique instructions to meet each student's specific needs are provided by the hybrid system, teachers can not only understand how this recommendation is made and how it would help their students, but machine learning algorithms can also ensure that the recommendation is predicted with the highest accuracy.

In addition to the superiority of the hybrid system, we demonstrated that AI-guided individualized language learning showed overall positive learning outcomes regardless of the system type (i.e., rule-based systems, machine learning systems, or hybrid). Therefore, we suggest that teachers and learners use each type of AI algorithm system according to their teaching and learning purposes. For example, machine learning AI systems generally tend to have more accurate predictive power than rule-based systems, whereas rule-based systems require humans to suggest interpretable rules for decision-making in algorithm systems. More specifically, although machine-learning AI systems have, on average, more significant treatment effects on enhancing language learning than rule-based AI systems, the latter have specific benefits, such as that we can understand and interpret their mechanism, which consists of rules predetermined by humans. For example, for applications such as Duolingo, which is based on a machine learning algorithm system, users want to receive the most appropriate lessons for their learning progress rather than to understand why certain lessons are given to them. Therefore, using machine learning AI systems in this case would be appropriate from learners' perspectives. On the other hand, in a school classroom setting, Connor's A2i software, which is based on rule-based systems, recommends that teachers provide specific lessons for certain student groups and, naturally, teachers can understand the rationale behind such recommendations and interpret them for their future instruction; therefore, rule-based systems can be more useful from teachers' perspectives.

Limitations and Directions for Future Research

In addition to the key insights offered by this study, our meta-analysis has some limitations. First, although we attempted to locate and review as many relevant studies as possible via diverse channels, there is the possibility of unidentified important studies (hence, the sign of publication bias from the pool of our between-group samples, as discussed above); some examples are unpublished papers with no significant, null, or even negative findings. Future meta-analytic endeavors should include a larger number of empirical studies, which would allow examining a wider range of moderators (e.g., language learning settings [e.g., laboratory, in-class, out-of-class] and AI hardware types [e.g., smartphones, computers, robots]). Second, given that the operationalization of AI is diverse among researchers, computer technicians, engineers, and laymen (e.g., Hechler et al., 2020), we acknowledge that our operationalization of AI-guided technology

(as per the moderator variable AI system type) can be challenged by some readers. Nonetheless, we strongly believe that our operationalization may be useful for categorizing an extant array of AI-guided technologies oriented toward individualized language learning and for providing pedagogical implications. Third, as mentioned in the Introduction, we did not consider language learning studies on chatbots, social robots, or AWE. However, this does not necessarily mean that these technologies cannot be used for individualized language learning (e.g., chatbots or robotics developed for general purposes can be applied to language learning contexts according to instructors' exclusive study design). Future meta-analysis studies may extend our endeavor by considering studies on the effects of such technologies geared toward individualized instruction. Finally, there was some imbalance in the number of primary studies associated with the 17 identified projects, with some projects (e.g., Project LISTEN, A2i) having been more extensively researched (i.e., using more samples) than others (e.g., ITS iSTART), although we attempted to resolve this imbalance by conducting a three-level meta-analysis.

Based on the results of the present study, we propose the following three directions for future research on AI-guided individualized language learning. First, with the ITS being the most widely categorized (e.g., Heilman et al., 2010; McNamara et al., 2006; Mills-Tetty et al., 2009; Poulsen, 2004), it seems necessary to examine in more detail the ITS aspects that facilitate language learning, ideally alongside qualitative data (e.g., examining user patterns). Second, the role of human instructors deserves closer evaluation within the wider array of AI technology employment, particularly in view of the existing skepticism regarding the AI's role in education (e.g., Chan & Tsi, 2023; Tao et al., 2019), including developing and designing curricula and managing the rules of learning material assignment. An exemplary empirical question would be whether the involvement of human instructors (or experts) would be more effective than the lack thereof in terms of utilizing the data obtained from the formative assessment for instruction afterward. Third, future research on this issue should refine the taxonomy of AI, based on which concrete implications can be provided for programmers and researchers of AI-based technology oriented toward individualized instruction.

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Appendix. Subgroup Analysis for Research Question 3

Table A and Figure A depict the results of the subgroup analysis for between-group samples.

Table A

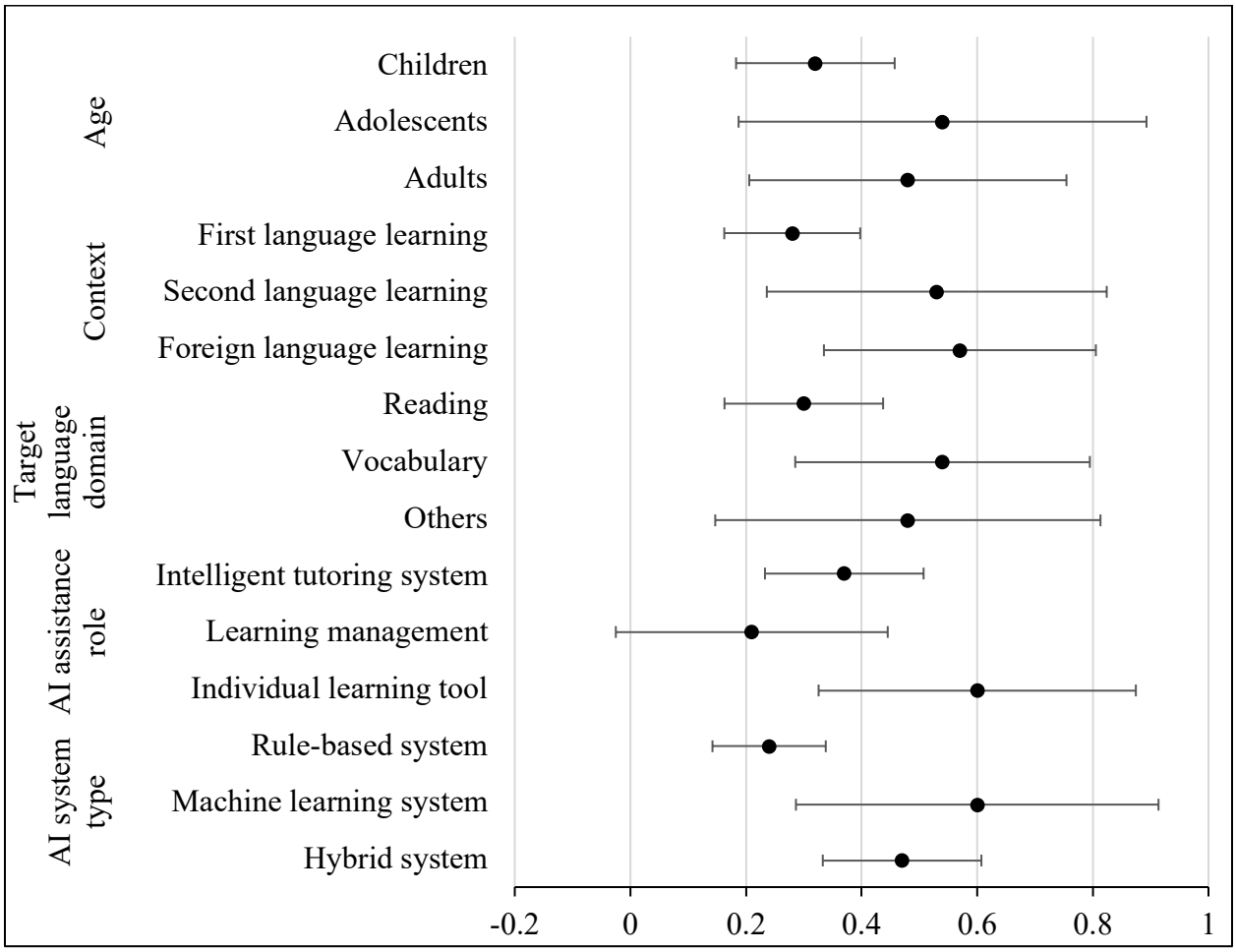
Results of Moderator Analysis (Subgroup Analysis) for Between-Group Samples (N = 6,020)

Independent Variable	Subgroup Analysis				
	# of ES	# of Sample	# of Project	Average ES	SE
Age					
Children	60	22	6	0.32***	(0.07)
Adolescents	5	4	3	0.54**	(0.18)
Adults	11	9	5	0.48***	(0.14)
Context					
First language learning	54	21	6	0.28***	(0.06)
Second language learning	8	2	2	0.53***	(0.15)
Foreign language learning	14	12	6	0.57***	(0.12)
Target language domain					
Reading	57	20	5	0.30***	(0.07)
Vocabulary	12	8	6	0.54***	(0.13)
Others	7	7	4	0.48**	(0.17)
AI assistance role					
Intelligent tutoring system	51	20	8	0.37***	(0.07)
Learning management	15	7	1	0.21	(0.12)
Individual learning tool	10	8	4	0.60***	(0.14)
AI system type					
Rule-based system	43	21	10	0.24***	(0.05)
Machine learning system	6	6	2	0.60***	(0.16)
Hybrid system	27	8	1	0.47***	(0.07)

Note. Four control variables, (a) duration of the instruction, (b) publication year, (c) study design, and (d) test reliability, were controlled for; none was statistically significant (*p*s > .05).
*** $p < .001$, ** $p < .01$

Figure A

Forest Plot for Between-Group Samples' Subgroup Analysis



Subgroup Analysis

Most importantly, for a general description, we found that the AI-guided individualized language learning had either small- ($0.20 \leq d < 0.50$) or medium-sized ($0.50 \leq d < 0.80$) effects on students' linguistic developments across various categorical values of five moderator variables. Given that there were no marginal effect sizes (i.e., $d < 0.20$), it was therefore found that the AI-guided individualized language learning were effective overall. With respect to statistical differences across moderator values, we could not identify any of them based on the 95% CIs (see Figure A).

For different age groups, the results indicated that the AI-guided individualized language learning had a medium-sized effect ($d = 0.54$, $SE = .18$) for adolescent students, but a small-sized effect for both children ($d = 0.32$, $SE = .07$) and adult learners ($d = 0.48$, $SE = .14$). Regarding language learning contexts, we found that there were medium-sized effects for both second language ($d = 0.53$, $SE = .15$) and foreign language learning ($d = 0.57$, $SE = .12$) contexts. However, there was only a small-sized effect for first language learning contexts ($d = 0.28$, $SE = .06$). As for target language domains, the average effect size was medium for improving students' vocabulary knowledge ($d = 0.54$, $SE = .13$), but they were small for improving students' reading skills ($d = 0.30$, $SE = .07$) and the other domains ($d = 0.48$, $SE = .17$).

With respect to the roles and types of AI systems, we identified that individual learning tools had a medium-sized average effect size ($d = 0.60$, $SE = .14$) and that the remaining two roles—intelligent tutoring systems ($d = 0.37$, $SE = .07$) and learning managements ($d = 0.21$, $SE = .12$)—had small-sized average effect sizes.

Per the AI system type, the results revealed that AI systems with machine learning systems had a medium-sized average effect size (d = 0.60, SE = .16), whereas the other types had small-sized average effect sizes, namely, the rule-based systems (d = 0.24, SE = .05) and hybrid systems (d = 0.47, SE = .07).

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